

# $\Sigma$ -check Interactive Proof

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## 1 Intro

- Sum check is an interactive protocol in which the prover proves the correct evaluation of the sum of a polynomial over the entire domain.
- The verifier also has the polynomial or in more formal terms, the oracle access to the polynomial  $g()$ , but wants to avoid calculating the whole sum and the smaller sums himself.

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$$H(g()) = \sum_{b_1 \in \{0,1\}} \sum_{b_2} \dots \sum_{b_v} g(b_1, b_2, \dots, b_v)$$

- $H$  is sum over all  $2^v$  possibilities, and verifier is polynomial time, but the prover is not.
- The last round requires the verifier to check against the oracle access given to it.
- Verifier has to evaluate a total of  $2 * v$  univariate polynomials (from  $g_1$  to  $g_v$ ) and 1  $v$ -variate polynomial (in the last round) to verify the prover's statement about  $C_1$ .
- If any other Field, let's say  $\mathbb{F}_5$  were used, then  $b_i \in \mathbb{F}_5$ ; the prover would have to evaluate  $g()$  at  $v$  possibilities and the verifier would have to check the sum over all 5 possibilities.  
 $g_j(X_j) = g_{j-1}(0) + g_{j-1}(1) + g_{j-1}(2) + g_{j-1}(3) + g_{j-1}(4)$

## 2 Protocol view

**Prover** ( $g(b_1, \dots, b_v)$ )

**Verifier** ( $g(b_1, \dots, b_v)$ )

**0-th Round**

$$C_1 = H(b_1, b_2, \dots, b_v)$$

$$\longrightarrow \xrightarrow{C_1} \longrightarrow$$

**1-st Round**

$$g_1(X_1) = H(X_1, b_2, \dots, b_v)$$

$$\longrightarrow \xrightarrow{g_1(X_1)} \longrightarrow$$

**Check**

1.  $g_1$  is univariate
2.  $C_1 = g_1(0) + g_1(1)$

**2-nd Round**

$$g_2(X_2) = H(r_1, X_2, \dots, b_v)$$

$$\longleftarrow \xleftarrow{r_1} \longleftarrow$$

$$\longrightarrow \xrightarrow{g_2(X_2)} \longrightarrow$$

$$r_1 \xleftarrow{R} \mathbb{F}_2$$

**Check**

1.  $g_2$  is univariate
2.  $g_1(r_1) = g_2(0) + g_2(1)$

**j-th Round**

$$g_j(X_j) = H(r_1, \dots, r_{j-1}, X_j, b_{j+1}, \dots, b_v)$$

$$\longleftarrow \xleftarrow{r_2} \longleftarrow$$

$$\vdots$$

$$\longrightarrow \xrightarrow{g_j(X_j)} \longrightarrow$$

$$r_2 \xleftarrow{R} \mathbb{F}_2$$

**Check**

1.  $g_j$  is univariate
2.  $g_{j-1}(r_{j-1}) = g_j(0) + g_j(1)$

$$\longleftarrow \xleftarrow{r_j} \longleftarrow$$

$$r_j \xleftarrow{R} \mathbb{F}_2$$

**$\nu$ -th Round**

$$g_\nu(X_\nu) = H(r_1, \dots, r_{\nu-1}, X_\nu)$$

$$\longrightarrow \xrightarrow{g_\nu(X_\nu)} \longrightarrow$$

**Check**

1.  $g_\nu$  is univariate
2.  $g_{\nu-1}(r_{\nu-1}) = g_\nu(0) + g_\nu(1)$

**Final check**

1.  $r_\nu \xleftarrow{R} \mathbb{F}_2$
2. Verifier checks using the oracle  $g()$   
 $g_\nu(r_\nu) = ? g(r_1, r_2, \dots, r_\nu)$
3. Verifier ACcEpts if check succeeds, REjEcts otherwise.